

(/)

Critical Material Recovery from E-waste



United States Energy Association: Critical Material Recovery from E-waste

Final Report

Subagreement No. 633-2023-004-01

Prepared by:

Battelle
505 King Avenue
Columbus, Ohio 43201

Submitted to:

United States Energy Association

Technical Point of Contact:
Dr. Morgan Evans
Lead Environmental Scientist
Battelle, Columbus, OH
Email: volker@battelle.org
Phone: 614-312-0415

Contractual Point of Contact:
Courtney Brooks
Contracts Manager
Battelle, Columbus, OH
Email: brooksc1@battelle.org
Phone: 614-424-5623

Technical Point of Contact:
Ms. Kathryn Johnson
Program Manager
Battelle, Columbus, OH
Email: johnsonk@battelle.org
Phone: 614-424-7035

August 30, 2023

This report was developed for the United States Energy Association (USEA) to describe the opportunity for recovering critical materials (CMs) from electronic waste (e-waste). The research summarized herein includes: 1) a review of the location and purpose of CMs in a selection of electronic devices, 2) descriptions of the current supply chain, recycling infrastructure and the challenges they pose in achieving e-waste circularity, and 3) an assessment of state-of-the art recovery technologies, their current scales, and the existing gaps. This report covers a subset of CMs that are most relevant to e-waste. This list includes aluminum, arsenic, gallium, indium, nickel, tantalum, tungsten, titanium, tin, platinum group

metals (PGMs - platinum, palladium, rhodium, ruthenium, osmium, and iridium), and rare earth elements (REEs - lanthanum, cerium, praseodymium, neodymium, promethium, samarium, europium, gadolinium, terbium, dysprosium, holmium, erbium, thulium, ytterbium, lutetium, scandium, and yttrium).

The review indicates that CMs are found in a range of concentrations in the major categories of e-waste covered in this report, which are wind turbines, solar panels, phones, computers, screens (cathode-ray tube [CRT], light-emitting diode [LED], or liquid-crystal display [LCD]), electric vehicle magnets, and medical devices. CMs are often in the form of magnets, semiconductors, and phosphors, but they can also be used for structural and electrical applications.

Overall, there is a lack of directly scaled and/or applicable technology for recovering CMs from e-waste. Promising laboratory-scale technologies should be studied at larger scales to validate them for industrial use. Existing industrial-scale recovery efforts should be expanded to secure future supplies. Additionally, there is a need to fund more research both in 1) technologies to recover CMs from specific components, such as transparent screens, with high concentrations of high-value materials, and 2) technologies / strategies for comprehensively recycling products that contain multiple valuable materials.

However, bridging gaps in CM recovery technologies alone is not sufficient to achieve circularity. A variety of strategies should be integrated with technological innovations to drive recovery efforts such as: better regulations, policy changes, connections between supply chain segments, end-of-life management, incentive structures around collection and recycling, incentives around better product design while incorporating R10 circular strategies, and stakeholder behavior changes.

In summary, this report illustrates that there is both a significant opportunity and a need for recovering CMs from e-waste through technological innovations and supply chain/recycling infrastructure improvements. CMs are vital for many modern technologies and defense applications, especially those related to clean energy. Recycling e-waste is a crucial strategy to enable a domestic market supply chain capable of producing a long-term supply of these elements.

Attachment	Size
 USEA633-2023-004-01_CMfromEwaste_FINAL_REPORT.pdf (https://usea.org/sites/default/files/USEA633-2023-004-01_CMfromEwaste_FINAL_REPORT.pdf)	1.29 MB

Energy Category: Coal (/energy-category/coal)

 [Share](#)

 (<https://twitter.com/share?url=https%3A//usea.org/publication/critical-material-recovery-e-waste&count=none&via=USEnergyAssn&text=Critical%20Material%20Recovery%20from%20E-waste&counturl=usea.org/publication/critical-material-recovery-e-waste>)

 (<https://www.facebook.com/sharer.php?u=https%3A//usea.org/publication/critical-material-recovery-e-waste&t=Critical%20Material%20Recovery%20from%20E-waste>)

[Print \(/print/4732?utm_source=PANTHEON_STRIPPED\)](#)

 [PDF \(/printpdf/publication/critical-material-recovery-e-waste?utm_source=PANTHEON_STRIPPED\)](#)

RELATED PROGRAM



(/program/building-consensus-carbon-capture-utilization-and-

storage-ccus-and-clean-energy-systems)

Building Consensus on Carbon Capture, Utilization and Storage (CCUS) and Clean Energy Systems

(/program/building-consensus-carbon-capture-utilization-and-storage-ccus-and-clean-energy-systems)

LATEST



(/article/usea-commemorates-34-years-us-georgia-relations-and-decades-energy-sector-

partnership)

USEA Commemorates 34 Years of U.S.-Georgia Relations and Decades of Energy Sector Partnership

(/article/usea-commemorates-34-years-us-georgia-relations-and-decades-energy-sector-partnership)

April 24, 2026

USEA

Explores
Grid



(https://usea.org/article/usea-explores-grid-modernization-ceba-ceo-rich-powell)

Modernization with CEBA CEO Rich Powell (/article/usea-explores-grid-modernization-ceba-ceo-rich-powell)

April 23, 2026



(https://usea.org/article/supporting-new-technologies-new-technologies-%E2%80%93-

collaboration-based-approach-smrs)

Supporting New Technologies with New Technologies – A Collaboration-Based Approach to SMRs

(/article/supporting-new-technologies-new-technologies-%E2%80%93-collaboration-based-approach-smrs)

April 3, 2026



(<https://usea.org/article/usea-statement-middle-east-stability-and-american-energy->

dominance)

USEA Statement on Middle East Stability and American Energy Dominance (/article/usea-statement-middle-east-stability-and-american-energy-dominance)

April 2, 2026

What will it take to
Finance American
SMR Projects in
Eastern Europe?

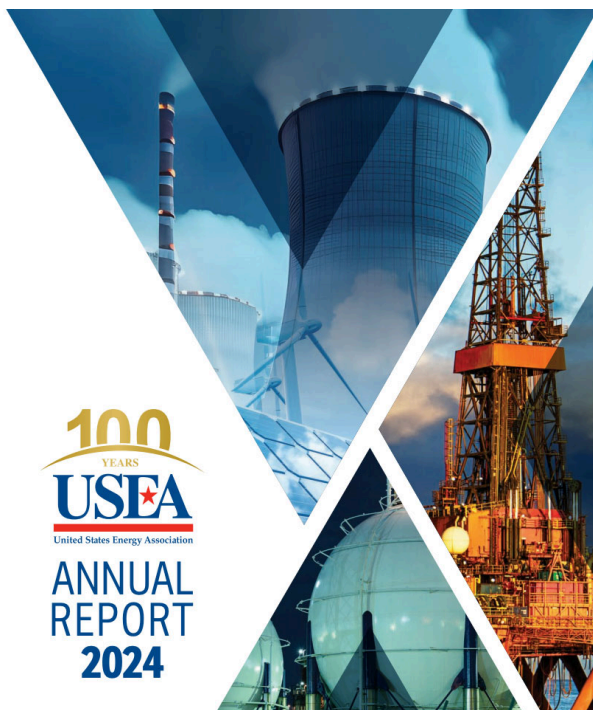


(/article/what-will-it-take-finance-american-smr-projects-eastern-europe)

(/article/what-will-it-take-finance-american-smr-projects-eastern-europe)

March 30, 2026

USEA ANNUAL REPORT



(<https://usea.org/publication/usea-2024-annual-report>)

ABOUT USEA

The United States Energy Association (USEA) is a nonprofit, apolitical, nonlobbying organization founded in 1924. USEA's mission has two pillars of equal importance. USEA serves as a resource, by convening energy stakeholders to share policy, scientific, and technological information to foster the advancement of the entire energy sector. Internationally, USEA promotes energy development by expanding access to safe, affordable, sustainable, and environmentally acceptable energy in partnership with the U.S. Government. [more \(https://usea.org/about-usea\)](https://usea.org/about-usea)

CONTACT US

United States Energy Association

1300 Pennsylvania Avenue, NW

Suite 550, Mailbox 142

Washington, DC 20004-3022

☎ (202) 312-1230

✉ reply@usea.org (<mailto:reply@usea.org>)



[Log In](#)